Field Hours

CSCE 655 Human-Centered Computing

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 40 minutes

Needfinding

Human-centered work starts with a person doing a task, not with a screen you want to build. Needfinding is the set of observations that justify a design. An interview is one tool. Watching the task is a better one when the person cannot see their own workarounds. They will tell you the official procedure and then skip three steps. The skip is the data.

Do not lead the witness. 'Would you like a dashboard' is a question about your idea, and polite people say yes. 'Walk me through the last time you filed this form' is a question about their life. Ask for a recent instance, not a generalization. 'I usually' is a story. 'Last Tuesday' is evidence.

Write what you saw, separate from what you concluded. A note that says 'the user is confused' is a conclusion. A note that says 'they clicked Save, nothing visible changed, they clicked it three more times' is an observation. Design from the observations. The conclusion can be wrong and the clicks remain.

A team project needs a shared set of notes. If each member interviewed a friend and kept the recording private, you do not have a sample. You have anecdotes. Put the raw notes where the team can read them before anyone draws a screen. The screen is a hypothesis about the notes.

Two questions. Only one of them can surprise you.

A leading question

Would a dashboard help

The answer is yes

You learned nothing

A recent instance

What happened Tuesday

They skipped three steps

The skip is the need

A clinic front desk

1. You ask how check-in works. The answer is a tidy five-step policy.

2. You watch one patient. The clerk copies a number from a card onto paper because the field will not take the dash the card prints.

3. The need is the field, or the card, not a new dashboard. You would not have heard it from the policy.

Problem 1. Four interviews all say they want 'more analytics'. What is missing before you build analytics?

Answer. A recent instance of a decision that failed for lack of a number. Without that observation, 'analytics' is a word they used to be helpful. Watch the decision or ask for the last time it happened.

Problem 2. Your notes say 'user hates the form'. Why is that note weak?

Answer. It is a conclusion. It does not say what they did. A usable note records the click, the pause, or the sentence. Another designer can disagree with 'hates' and still use the click.

Lecture 2 · 40 minutes

Prototypes you can throw away

A prototype is a way to be wrong cheaply. Paper, a slide, or a clickable fake is enough when the question is whether a person understands the control. Production code is the wrong material for that question, because you will refuse to throw it away. The cost of the prototype should be low enough that a 'no' from a user is a relief, not a loss.

Match the fidelity to the question. A visual question needs a picture at the size the person will see. A workflow question needs the steps in order, even if they are boxes on paper. A performance question needs data and a timer, not a picture. A pretty picture does not answer a workflow, and a workflow sketch does not answer whether the type is readable.

Information structure is the part you can prototype before color. What are the objects, how are they grouped, what is the name of each group. If the person cannot say which group a task lives in, the structure is wrong and the typeface will not save it. Draw the groups as boxes. Ask the person to place a recent task in a box. Move the boxes. This is still a prototype.

The team should be able to say what the prototype is not. It is not a commitment to the database. It is not a brand. Write the non-goals on the back of the paper. A stakeholder who sees a polished prototype will hear a promise. Use a rougher one if you are not ready to keep it.

One question, one cheap artifact, then a decision. The artifact is allowed to die.

Name the question -> Build the cheapest fake -> Watch someone use it -> Keep or discard

Where the dash goes

1. Question: will a clerk enter the card number correctly if the field shows the dashes.

2. Prototype: a paper form with the boxes printed, and a pencil. Not a database.

3. Two clerks try a real card. You watch where they pause. Then you throw the paper away and keep the decision.

Problem 1. The team spends three weeks on a prototype because the spacing is not final. What did they forget?

Answer. The question the prototype was for. Spacing is a later question. Three weeks means a no is now expensive, so they will not hear it. Cut the prototype back to the question.

Problem 2. A stakeholder treats the paper sketch as a delivered design. What should have been written on it?

Answer. The question it answers and the non-goals, including that the data model is not decided. A sketch with no label is read as a promise.

Lecture 3 · 40 minutes

A test with a success criterion

A usability test is a person doing a task while you watch. It is not a focus group, and it is not you clicking through your own prototype. You write the task in the person's language, without naming the control. 'Check in the patient on this card' is a task. 'Click the dashed field' is a hint. Read the task aloud the same way every time.

Success is defined before the session. For example: the clerk enters the number so it matches the card, in under a minute, without asking you what to do. If you decide after the session whether it went well, you will decide it went well. Write the criterion on the same page as the task.

Think-aloud is optional and changes the task. Some people cannot talk and type. Silence plus what they click is data. Help only after they are stuck past a threshold you set, and count that session as helped. A success you coached is not a success.

Five people will find the obvious breaks. They will not estimate a percentage in the population. Do not report '80 percent of users' from four out of five participants. Report what happened: three could not find the field, one entered the dashes, one finished inside the minute. The next design change is about those observations.

A session. The criterion is written before the person arrives.

1. Read the task

2. Stay quiet

3. Note what they do

4. Score against the criterion

A one-minute check-in

1. Task: check this patient in using the card. Criterion: the stored number matches the card, under one minute, no help.

2. The first clerk searches the header for a while and finishes in two minutes. That is a failure against the criterion, even though they eventually succeed.

3. You do not move the goalposts. You note where they searched.

Problem 1. You tell the participant 'the field is on the left' after ten seconds. Can you count a success?

Answer. No. You coached the task. Mark the session helped. The ten seconds is the observation: they had not found it yet.

Problem 2. Four of five people succeed. Why is '80 percent usability' a bad sentence in the report?

Answer. Five people is not a sample of the population. The honest sentence is four of these five

completed the task, plus what the fifth did. A percentage implies a measurement you did not make.

Lecture 4 · 40 minutes

Visual design, structure, and access

Visual design is how the eye knows what matters. Hierarchy is size, weight, and position used consistently. If everything is bold, nothing is. Alignment is a grid the person never names and immediately feels when it breaks. Contrast is what makes text readable in the room you have, not on your laptop in a dark office. These are constraints on the prototype, and they are testable: can the person find the primary action without being told.

Information structure is which objects exist and how they nest. A patient, a visit, a bill. If the bill is drawn as if it were a visit, the person will put money in the wrong object. The structure is the design. Color is not. Fix the structure in the boxes from lecture two before you pick a palette.

Accessibility is a constraint on the same objects. A person who does not see the screen still has to complete the task. That means the structure is in the text and the controls, not only in the position of a colored box. Labels are visible and attached to fields. The action is reachable without a precise gesture. Contrast meets a number you can measure. You do not get to call the pretty version finished and the accessible version a later coat of paint. The paint is the only version some people can use.

Test with the constraint in the room. A keyboard-only pass by you, once, will find the controls that are not reachable. It is not a substitute for a person who uses a screen reader, and it is more honest than a claim you did not try. Write the access criterion next to the time criterion: the task can be finished without a pointer, or it cannot, and you watched.

What the person is actually parsing. Decoration is the bottom layer and the last one you earn.

A task they can finish

Objects and groups

Labels and contrast

Decoration

A primary action

1. The screen has four buttons of equal size and equal color. The clerk asks which one saves the check-in.

2. You make Save the only filled button and put it where the other primary actions live. The next clerk does not ask.

3. The change was hierarchy, not a new feature. The structure of the task was already known.

Problem 1. Error text is red on a dark red button. The clerk with typical vision can read it. Why is the design still wrong?

Answer. Contrast is too low for people who do not have the clerk's vision, and color is the only carrier of the error. Use text that says what to fix, and a contrast you can measure, so color is not the only signal.

Problem 2. The team plans to 'add accessibility at the end'. What decision are they postponing?

Answer. Whether the structure and the actions are in the interface a keyboard and a reader can reach. Those are design choices. Left until the end, they become a rewrite, or they are skipped.

Lecture 5 · 45 minutes

One test, and what you change

Run one test on a paper or clickable prototype of the check-in field. Recruit three people who do the job or who can stand in without being on the team. Team members are a last resort and a biased one: they already know where the field is. Give them a real card. Use the task and the criterion from lecture three. Include the access pass from lecture four on at least one session: no pointer.

Record observations, not a highlight reel. What they touched first. Where they paused. Whether the stored number matched. Whether they needed help. Separate those notes from the change you want to make. The change is a hypothesis about the notes. 'Make it cleaner' is not a hypothesis. 'Show the dashes in the field so the card and the field match' is a hypothesis, and it is the one the clinic story was pointing at.

Change the prototype, not a document about the prototype. Then run one more session, even a single person, on the change. A redesign that was not retested is a guess you got attached to. The second session is allowed to be short. It is not allowed to be imaginary.

Write the team report in this order: the task, the criterion, what happened, what you changed, what happened after. A report that starts with a screenshot and ends with a claim of delight is a pitch. This course grades the order. The project is done when the second session's notes are in the report, including a failure if you still have one.

The whole project in one strip. The second session is part of the work, not a prize.

Watch a real task -> Prototype one change -> Test against a criterion -> Test the change once more

Dashes, before and after

1. Before: two of three clerks pause and one copies the number onto paper. Criterion failed for those two if the minute included the pause and the mismatch.

2. Change: the field prints the dashes in the same places as the card. Nothing else moves.

3. After: the next clerk enters the number without asking. One session is not a population. It is evidence the change is worth keeping while you watch a fourth person.

Problem 1. The second session fails the criterion. What do you put in the report?

Answer. The failure, the observation, and the next change you would test. A report that hides the second session is a pitch. The course asked for the notes.

Problem 2. Only teammates are available, and they helped design the field. How do you label their sessions?

Answer. As biased. They already know the answer to the only question the test can ask. Say so at the top of the notes. Do not report their success as a user success.

Lecture 6 · 180 minutes

A person takes time

The earlier weeks found a need and tested a prototype. This week is a quantitative model you can use before the test. Fitts' law says the time to point at a target grows with the distance and shrinks with the size. A small button far from the current pointer is slower and less accurate. The law is a log of distance over width, plus a constant. You do not need the fitted constants to use the shape of it.

Hick's law says the time to choose grows with the log of the number of equally likely options. A menu of forty peer items is a tax. Grouping, or a default, is how you stop charging it. The law does not excuse hiding the option people need. It tells you the cost of laying them out as peers.

Working memory holds a few chunks. A form that asks for a code, a date, and a reason, and then clears them when a validation fails, spends the user's memory for nothing. Keep the values. The error belongs next to the field.

These models are hypotheses about the interface, tested by the usability session. If the model says the button is fine and three people miss it, believe the people.

Two costs you can see.

Pointing

Far and small is slow

The edge of the screen is a large target

Choosing

More peer options take longer

A default removes the choice

A 12-pixel button

1. The primary action is 12 pixels tall, in the corner opposite the pointer's usual rest, beside a 40-pixel destructive action.

2. Fitts says the destructive action is the easier target. People will hit it by mistake.

3. Swap the sizes, or move the destructive action away and confirm it. The model told you before the test. The test will tell you if you were wrong about which action is primary.

Problem 1. You add a confirmation dialog to every action, including the common one, to 'prevent mistakes.' What cost did you add?

Answer. A choice and an extra pointing step on the frequent path. Hick and Fitts both go up on the action people came to perform. Confirm the destructive rare action. Do not confirm the primary one.

Problem 2. A validation error clears the form. Why is that a memory bug and not only a manners bug?

Answer. The user must hold every field in working memory to retype them, and they will misremember one. Keep the text. Show the error beside the field that failed.

Lecture 7 · 180 minutes

Structure

Information architecture is which things sit together and what they are called. A label is wrong when two people looking for the same thing choose different labels and only one of them is in the menu. Card sorting, even with five people and paper, finds that disagreement before you build the menu.

Hierarchy earns itself when the branches match the user's task. An org-chart hierarchy matches the company and often fails the task. Group by the job the user is doing this minute.

Search is not a substitute for a structure nobody can browse, unless you have measured that people can name the thing. A person who does not know the term 'prior authorization' will not type it. The label has to contain a word they have.

Wayfinding tells people where they are and how to go back. A page with no name and a logo that does not go home is a trap. The back control must undo the last step the user thinks they took, not a step in your implementation.

A structure you can test.

Labels in the user's words

Groups by task

A name for the current page

A way back

Two names for one form

1. Staff call the form 'check-in.' Patients call it 'sign in' or 'the clipboard.' The menu says 'encounter initiation.'

2. Three people in a hallway test cannot find it. They can find 'sign in.'

3. Rename the menu. The database table can keep its name. The label is not the schema.

Problem 1. Why can a search box fail even when the page exists?

Answer. The user does not know the words in the page. Search matches strings. If the label and the query share no words, the page is invisible. Browseable task groups, using their words, are the backup.

Problem 2. The logo does not navigate home and there is no other home control. What is the wayfinding failure?

Answer. People cannot reset the task without the browser's back button, which may leave your application in a state you did not design. Give them a control that returns to a known page.

Lecture 8 · 180 minutes

How you evaluate

A usability test with a person and a task is the method you already used. A heuristic review is an expert reading against a list: visibility of status, match to the user's language, control and freedom, consistency, error prevention. It finds cheap problems and misses problems only a real task reveals. Do the heuristic pass before you spend the participant's time, not instead of it.

Five participants find a large fraction of the obvious problems in a small interface, and they do not find rare ones. The number five is a rule of thumb from a simple model, not a law. If the first three people fail in three different ways, you are not done at five. If they all fail the same way, fix it before you recruit the next five.

A survey after the fact measures opinion. It does not measure whether they could do the task. Ask for the time and the success before you ask whether they liked it. People like interfaces they failed at, and dislike interfaces they completed, often enough that the scores are not the result.

Comparative tests need a fair task. Changing the task between version A and version B measures the task. Keep the task, change the interface.

Two evaluations.

Heuristic review

An expert and a list

Fast, no participant

Misses task failures

A session

A person and a task

Slow, specific

Shows the actual miss

Five people, one button

1. All five fail to find the export control, which sits in a menu named 'tools.'

2. You do not need a sixth person to know the label is wrong. Fix it and run a smaller confirmation.

3. A survey that asked 'was the tools menu useful' would have returned opinions. The failures were the data.

Problem 1. You run a heuristic review and find no issues, and the first participant cannot complete the task. Which result do you trust?

Answer. The participant. The heuristic list did not contain their task. Watch the session, fix the task failure, and then add the missing check to your own list.

Problem 2. Why is 'three out of five liked it' not a success criterion?

Answer. Liking is not task success. Define success as completing the task in the time you named, without help. Report liking separately if you measured it, and do not let it replace the count of failures.

Lecture 9 · 180 minutes

Reading a small study

Five people are not a p-value. Report counts. Three of five failed. The times were 40, 55, 70, 90, and 200 seconds. The 200 is a person you should describe, not a number to delete because it ruins the average. The median is more honest than the mean on a set this small with one long tail.

Do not compute a statistical test you cannot interpret and then write 'significant.' A claim of a difference between two versions with five people each is usually a description, not a detection. Say what happened.

Consent and harm belong here. Tell the person the task, that they can stop, and that you are testing the interface and not them. Do not record more than you said you would. Delete the recording when the note is written, if that was the agreement.

A single quoted sentence from a participant can carry the finding, next to the count. 'I thought check-in was the clipboard' plus three failures is a result. The sentence alone is an anecdote. The count alone is a table. Keep both.

What to report.

How many people, how many failures

The times, including the long one

One sentence they said

No test you cannot explain

Five times

1. Times in seconds: 40, 55, 70, 90, 200. The mean is 91. The median is 70.

2. The person at 200 gave up and used a paper form. That is a failure, not a slow success.

3. Report 4 of 5 completed, median 70, one abandonment. The mean hid the abandonment if you only plotted it.

Problem 1. You drop the 200-second session because it was 'an outlier.' What did you hide?

Answer. A failure mode. Outliers on five people are often the finding. Describe the session. Do not delete it to tighten the average.

Problem 2. Both versions have five people and version B is 10 seconds faster on the median. What may you claim?

Answer. That in this sample the median was 10 seconds lower. You may not claim you have detected a general advantage. The sample is a pilot. A larger difference, or a difference in success counts, is the thing worth acting on.

Lecture 10 · 180 minutes

Access

The earlier lecture put contrast and a primary action on the page. This week is the rest of access. A person who

does not use a pointer needs every action from the keyboard, in an order that follows the task, with a visible focus. A person who uses a screen reader needs a name on every control. A button whose only content is an icon has no name unless you give it one.

Captions and text alternatives are the same idea for other senses. An error communicated only by color is invisible to someone who does not see that color. Add the text.

Zoom must not clip the primary action. A layout that locks the font size will be overridden badly. Let the text grow and let the page reflow.

Access is a requirement in the sense of the software course: a sentence a test can fail. 'Works with a keyboard' fails when you cannot reach the export control without a pointer. Put that test in the suite. Do not leave it as a final-week surprise.

A keyboard pass.

1. Start at the top

2. Tab in task order

3. See the focus

4. Activate without a pointer

5. Hear a name, not 'button'

An icon with no name

1. The delete control is a trash-can image with no text.

2. A screen reader says 'button' or nothing useful. The keyboard user can focus it and does not know what it will do if the focus style was removed.

3. Add an accessible name and a visible focus outline. The visual design can keep the icon. The name is not optional metadata.

Problem 1. The error is a red border and no text. Who cannot use it, even among people who see the screen?

Answer. Anyone who does not perceive that red, and anyone who does not know the border means error rather than focus. Add the words. Color can stay as a duplicate signal.

Problem 2. Tab order follows the visual order of a two-column layout that was implemented as one column in the markup, then moved with styles. What does the keyboard user experience?

Answer. Focus jumps in the markup order, not the visual order. They tab to a field that is not the next one on the screen. Put the markup in task order, or set the order deliberately and test it.

Lecture 11 · 180 minutes

Studies and the people in them

A study uses someone's time and their behavior. The minimum standard is informed consent, a way to stop, data minimization, and a purpose you told them. A usability test of a school project is still a study. 'It is just a class' is

not a waiver.

Do not test on people who cannot refuse you: an employee who reports to you, a patient waiting for care, a student whose grade you control, without a path that does not involve you. The power is the ethical problem, not the questionnaire.

Store the notes without the names if the names are not the data. A video of a person's face is identifiable. Prefer notes taken in the session if you do not need the video. If you need it, say so before they start.

The finding should be allowed to be 'do not build this.' A study that can only end in shipping is a demonstration. The clinic front desk may not need your application. That is a successful study if you learned it before they had to live with the application.

Before you recruit.

What you will ask them to do

That they can stop

What you will keep

Who must not feel obliged

A front desk during clinic hours

1. You want to watch staff use a prototype. The waiting room is full. They cannot stop without abandoning a patient.

2. That is not consent. Schedule a time when stopping is real, or do not run the session there.

3. A hallway card sort with a volunteer who knows they can walk away is a different study. Use that if you cannot get the real setting ethically.

Problem 1. You record the session 'in case,' and the consent form said notes only. What do you do with the file?

Answer. Delete it. You do not have consent to keep it. The notes you said you would take are the data. A useful recording you were not allowed to make is not a reason to amend the agreement after the fact.

Problem 2. The study shows the tool slows the task relative to paper. What is a valid conclusion?

Answer. Do not replace the paper, or change the tool and test again. Shipping it because the semester ends is not a finding. The slowdown is the finding.

Lecture 12 · 180 minutes

A design you can iterate

The project is one task, two versions, and a session. Version one is the prototype from the early weeks. You run the session, count the failures, change one thing that the failures named, and run a confirmation. The earlier lecture's retest is this, with the models and the access checks included.

Change one thing. If you redesign the whole screen, you do not know what fixed the failure. The software course

said the same about diffs. It is the same idea.

Include the keyboard pass and one access requirement in the confirmation. A visual fix that removes focus is not a fix.

Write a page: the task, the count, the change, the second count. That page is the deliverable. A slide of screenshots is not.

Two versions.

A task and a success rule -> Five sessions or fewer -> One change, tied to a failure -> A confirmation session

Dashes, before and after

1. Before: three of four people type the id without dashes and the field rejects them. Times are long because they retype.

2. The change: accept the digits with or without dashes, and show the canonical form. One change.

3. After: four of four enter the id. If a new failure appears, you write it down and do not pretend the confirmation was clean.

Problem 1. You change the label, the order, and the button in response to one failure. What can the confirmation tell you?

Answer. Only that the bundle differs. You cannot attribute the new result to the label. Change the one thing the failure named, or call the session a new pilot rather than a confirmation.

Problem 2. The confirmation removes the text alternative so the layout matches a screenshot. What requirement did you break?

Answer. The access requirement that the control has a name. The visual match is not the success criterion. Put the name back before you call the iteration done.

Lecture 13 · 180 minutes

The interface of the capstone

The capstone has a person who is not you. That phase is this course. Before you build the screen, write the task, the success criterion, and the one access sentence. Then prototype on paper. A paper session is cheaper than a session of a finished wrong screen.

The security course will forbid some convenient designs: a session in the URL, an error that reveals whether the account exists, a report tool that takes a raw path. Those constraints are part of the interface. A usable design that fails them is not a candidate. Find a design that a person can finish and that the threat model allows.

The database course will ask for a transaction around the action the button performs. The button's label should be the action the transaction actually commits. A button named Save that also sends a message and charges a card is three tasks pretending to be one.

Bring the one-page study plan to the capstone hours. Do not invent the user test on the last day.

Before the screen.

The task in one sentence

Success, timed or counted

The access sentence

The security constraint the screen must not break

A button with three effects

1. The button says Save. It saves the record, emails a third party, and submits a charge.

2. A person who wanted to save a draft has sent the email and the charge. The label lied.

3. Split the actions, or rename the button to the real effect and confirm the costly part. The transaction boundaries should match the labels.

Problem 1. The usable design puts the account number in the URL so the back button works. What do you change?

Answer. The navigation, not the security rule. Keep the account number out of the URL and the logs, and give the back control a real history. A usable leak is still a leak.

Problem 2. You have no participant yet and the interface is already coded. What do you do this week?

Answer. Paper or a clickable sketch, one task, one person who can refuse. You will still learn whether the label is wrong. Waiting for the finished UI wastes the iteration.

Lecture 14 · 180 minutes

What you owe the person using it

You should be able to run a small study, report counts without theater, name a pointing or labeling problem with a model, and check a keyboard path. You should also be able to stop a study you should not be running. That is the course.

The hours include one session and one revision, not a reread. The notes are the checklist. The session is the evidence.

When the capstone's last phase says a person who is not you, this is the method. Five tasks, a way to stop, a change tied to a failure, and a sentence about what you refused to change.

The work is original teaching of standard human-computer practice. It is not a copy of a studio course. The standard is the part that matters: watch a person, count what happened, change one thing.

The close.

A task

A count

One change

A person who could have stopped

A session you stop

1. The participant is frustrated and asks to stop on task two of five.

2. You stop. You do not coax a complete dataset. The note says they stopped, and why.

3. That session still counts as information. It does not count as a success. You do not replace their tasks with your own performance of the UI.

Problem 1. What do you report if one of five people stops and the other four succeed?

Answer. Four successes, one stop, and the reason they gave. You do not report five sessions completed. The stop is part of the result.

Problem 2. Name one model and one access check you apply before recruiting the next session.

Answer. Fitts or Hick on the primary control, and a keyboard pass that reaches it with a visible focus and a name. Then recruit. A session spent discovering that the button has no name is a session you did not have to spend.